

KING SALMON, AK, USA

WMO: 703260

Lat: 58.683N

Lon: 156.656W

Elev: 67

StdP: 14.66

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25503

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -24.1	(c) -17.7	(d) -28.2	(e) 1.1	(f) -21.7	(g) -22.4	(h) 1.6	(i) -16.1	(j) 35.5	(k) 38.7	(l) 30.9	(m) 37.8	(n) 4.3	(o) 10	(p) 0.769

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.3	73.1	59.5	69.4	57.8	66.1	56.2	61.2	70.2	59.3	66.7	57.5	63.7	9.4	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.5	70.6	63.3	55.9	66.6	61.6	54.3	63.0	60.0	27.1	70.1	25.9	66.7	24.8	63.9	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.8	24.3	20.5	-27.5	79.0	8.1	4.5	-33.3	82.3	-38.0	84.9	-42.6	87.5	-48.5	90.8		
			-26.4	64.1	6.5	3.5	-31.1	66.6	-34.9	68.7	-38.6	70.6	-43.4	73.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.0	16.6	22.9	22.6	35.1	44.8	52.1	55.9	55.1	48.4	36.0	24.5	17.9		
	DBStd	18.07	18.37	14.90	13.25	7.94	5.51	4.51	4.28	4.24	5.50	9.55	13.20	16.91		
	HDD50	5592	1037	759	848	446	174	26	4	7	91	438	766	996		
	HDD65	10569	1502	1179	1313	896	625	387	284	307	499	900	1216	1461		
	CDD50	500	0	0	0	0	14	89	187	166	41	3	0	0		
	CDD65	3	0	0	0	0	0	0	2	1	0	0	0	0		
	CDH74	85	0	0	0	0	6	24	43	12	0	0	0	0		
	CDH80	8	0	0	0	0	0	2	5	1	0	0	0	0		
(14) Wind	WSAvg	9.2	9.8	10.2	9.7	9.9	9.6	8.3	8.2	8.3	8.6	9.2	9.3	9.2		
(15) Precipitation	PrecAvg	20.5	1.0	0.8	0.9	1.0	1.4	1.7	2.3	3.1	3.0	2.2	1.6	1.4		
	PrecMax	32.8	2.4	2.2	3.4	3.0	3.3	3.3	5.0	5.7	7.3	4.6	4.8	3.7		
	PrecMin	13.2	0.2	0.1	0.0	0.2	0.2	0.2	0.7	1.1	0.9	0.0	0.0	0.2		
	PrecStd	3.7	0.6	0.6	0.7	0.6	0.7	0.7	0.9	1.1	1.5	1.0	0.9	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.1	45.3	48.5	58.3	72.3	78.2	79.1	75.4	66.6	56.8	50.1	46.4		
		MCWB	41.2	40.0	40.9	46.2	53.9	59.4	63.7	62.0	56.8	49.9	45.6	41.2		
	2%	DB	42.5	42.6	44.4	53.2	65.8	71.8	73.4	70.3	62.4	54.0	47.3	43.2		
		MCWB	38.6	38.2	39.0	43.4	51.2	57.1	61.1	59.5	54.1	48.4	43.4	39.2		
	5%	DB	40.7	40.9	41.5	50.1	60.4	67.3	69.2	67.0	59.5	51.4	44.5	41.1		
		MCWB	37.3	37.0	37.1	42.0	48.6	55.4	58.7	58.1	52.9	46.8	41.3	37.7		
	10%	DB	38.9	39.0	39.2	47.2	56.3	63.2	65.5	63.8	57.2	49.1	41.3	38.8		
		MCWB	36.1	35.6	35.4	40.4	46.6	53.3	56.8	56.6	52.0	45.0	38.7	36.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.6	41.0	41.8	47.9	55.7	61.2	64.9	64.0	58.9	51.7	46.5	42.1		
		MCDB	45.4	44.1	46.9	56.4	70.3	74.3	76.8	72.8	63.4	55.0	49.1	45.3		
	2%	WB	39.2	38.8	39.4	44.2	51.9	58.3	61.8	60.8	55.6	49.4	44.0	39.6		
		MCDB	41.9	42.0	43.7	51.9	63.6	69.6	72.0	67.8	59.9	52.9	46.8	42.4		
	5%	WB	37.6	37.3	37.5	42.3	49.5	56.1	59.6	59.1	53.9	47.3	41.6	37.9		
		MCDB	40.2	40.5	41.1	49.3	59.2	65.5	67.3	65.2	57.9	50.7	44.4	40.7		
	10%	WB	36.1	35.7	35.6	40.8	47.5	54.0	57.7	57.5	52.4	45.7	38.8	36.3		
		MCDB	38.7	38.6	38.7	46.8	55.2	61.8	64.0	62.3	56.3	49.0	41.1	38.8		
(35) Mean Daily Temperature Range	MDBR		13.8	14.5	16.6	16.8	18.5	18.4	15.3	15.0	15.2	14.6	13.6	14.2		
	5% DB	MCDBR	9.5	12.1	15.2	21.7	29.3	28.3	24.6	22.2	18.9	14.1	11.8	10.9		
		MCWBR	7.4	9.3	11.0	13.6	16.2	15.0	13.3	12.5	11.6	10.0	9.5	8.7		
	5% WB	MCDBR	9.6	11.8	13.8	19.7	25.2	25.1	22.5	18.7	15.4	12.5	11.9	10.5		
		MCWBR	7.6	9.4	10.3	12.8	14.4	13.9	12.6	11.0	10.0	9.4	10.1	8.6		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.307	0.317	0.361	0.384	0.377	0.371	0.365	0.334	0.307	0.307	0.266		
	taud		2.162	2.223	2.255	2.255	2.267	2.365	2.427	2.445	2.502	2.436	2.073	2.016		
	Ebn at Noon		176	225	261	267	268	272	271	265	257	228	145	122		
	Edh at Noon		15	23	30	36	38	35	33	30	24	19	16	12		
(44) All-Sky Solar Radiation	RadAvg		139	342	875	1305	1572	1660	1371	1136	810	437	186	84		
	RadStd		20	53	123	138	156	159	146	131	97	45	24	8		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	N/A	+5.09	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page